

# Computer Vision Course Review

## 计算机视觉课程复习问答 (全中英对照精排版)

### Week1-Image-Processing-Basics (Thursday)

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#### 问题 1 / Question 1:

*You have a noisy color image and you want to detect edges. Which processing order is better and why?*

你有一张带噪声的彩色图像，想要检测边缘。哪种处理顺序更好，为什么？

- a. Convert to grayscale → Sharpen → Smooth → Edge Detection (转换为灰度 → 锐化 → 平滑 → 边缘检测)
- b. Convert to grayscale → Smooth → Edge Detection → Sharpen (转换为灰度 → 平滑 → 边缘检测 → 锐化)

**答案 / Answer: b. Convert to grayscale → Smooth → Edge Detection → Sharpen**

#### 解析 / Explanation:

[EN] Edge detection relies on calculating image derivatives (gradients), which act as high-pass filters and are extremely sensitive to high-frequency noise. Therefore, you must apply a smoothing filter (like a Gaussian filter) *before* edge detection to suppress the noise. Sharpening should only happen at the very end (if at all), because sharpening a noisy image before smoothing or edge detection would severely amplify the noise and create false edges.

[ZH] 边缘检测依赖于计算图像导数（梯度），这种操作类似于高通滤波器，对高频噪声极其敏感。因此，必须在边缘检测之前应用平滑滤波器（如高斯滤波）来抑制噪声。锐化应该放在最后（如果需要的话），因为在平滑或边缘检测之前对噪声图像进行锐化会严重放大噪声并产生虚假边缘。

#### 问题 2 / Question 2:

*Explain why median filter preserves edges better than mean filter. Give an example scenario where median filter is preferable.*

解释为什么中值滤波器比均值滤波器更能保留边缘。给出一个中值滤波器更适用的场景示例。

**答案 / Answer: Median filter doesn't create new average values / 中值滤波器不会产生新的算术平均像素值**

**解析 / Explanation:**

[EN] A mean filter calculates the arithmetic average of a local neighborhood, which artificially blends pixels on both sides of an edge, causing blurring. A median filter simply selects the middle value from the sorted neighborhood pixels. Because it does not create new pixel values through arithmetic averaging, it can remove extreme outliers while keeping the sharp step-transitions of edges intact.

[ZH] 均值滤波器计算局部邻域的算术平均值，这人为地混合了边缘两侧的像素，导致边缘模糊。中值滤波器仅从排序后的邻域像素中选择中间值。因为它不通过算术平均产生新的像素值，所以它可以去除极端的异常值，同时保持边缘的锐利阶跃过渡完好无损。

[EN] **Scenario:** Removing **Salt-and-Pepper noise** (impulse noise) from an image.

[ZH] **场景示例:** 去除图像中的**椒盐噪声**（脉冲噪声）。

## 问题 1 / Problem 1:

*Design a filter/pipeline to remove noise while preserving edges for an image corrupted by mixed noise (Gaussian + salt-and-pepper). How would you adapt parameters for smooth vs edge regions?*

设计一个滤波器/处理流程，用于去除被混合噪声（高斯噪声 + 椒盐噪声）污染的图像，同时保留边缘。如何针对平滑区域与边缘区域调整参数？

**答案 / Answer: Adaptive Pipeline: Median Filter → Bilateral Filter**

**解析 / Explanation:**

[EN] To handle mixed noise, a two-stage approach is most effective: 1. **Remove Salt-and-Pepper:** Use a **Median Filter** first. It is highly effective at eliminating impulse noise without blurring edges. 2. **Remove Gaussian Noise & Preserve Edges:** Use a **Bilateral Filter**. Unlike a Gaussian blur that treats all pixels identically, a Bilateral filter considers both spatial distance and intensity similarity, smoothing smooth regions

while protecting sharp intensity jumps (edges). **Parameter Adaptation:** In smooth regions, increase the range (intensity) variance of the Bilateral filter to perform stronger smoothing. In edge-heavy regions, decrease the range variance to ensure the filter does not smooth across the intensity discontinuity.

[ZH] 为了处理混合噪声，采用两阶段处理流程最有效：1. **去除椒盐噪声：**首先使用**中值滤波器**。它能极其有效地消除脉冲噪声且不会模糊边缘。2. **去除高斯噪声并保留边缘：**使用**双边滤波器**。与同等对待所有像素的高斯模糊不同，双边滤波器同时考虑空间距离和强度相似性，在平滑区域进行平滑的同时保护剧烈的强度跳变（边缘）。**参数调整：**在平滑区域，增大双边滤波器的值域（强度）方差以进行更强的平滑；在边缘密集的区域，减小值域方差，确保滤波器不会跨越强度不连续处进行平滑。

## 问题 2 / Problem 2:

*Design a kernel to detect diagonal edges. How can you extend it to detect edges in multiple directions simultaneously?*

设计一个检测对角边缘的核。如何将其扩展以同时检测多个方向的边缘？

**答案 / Answer: Use Prewitt or Sobel operator kernels for diagonals. Extend using a filter bank.**

### 解析 / Explanation:

[EN] A diagonal edge detector can be designed by rotating standard Sobel/Prewitt operators by 45 degrees. For example, a  $3 \times 3$  Prewitt diagonal kernel is:

$$D_1 = \begin{bmatrix} -1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

**Extension:** To detect multiple directions simultaneously, create a **filter bank** consisting of a set of kernels corresponding to various orientations (e.g.,  $0^\circ, 45^\circ, 90^\circ, 135^\circ$ ). You can convolve the input image with each kernel and then take the maximum response at each pixel location or compute the vector sum of gradients to obtain a comprehensive edge map.

[ZH] 可以通过将标准的 Sobel/Prewitt 算子旋转 45 度来设计对角边缘检测器。例如，

一个  $3 \times 3$  的 Prewitt 对角核如下：

$$D_1 = \begin{bmatrix} -1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

**扩展：**要同时检测多个方向，可以创建一个由对应于不同方向（如  $0^\circ, 45^\circ, 90^\circ, 135^\circ$ ）的核组成的**滤波器组**。你可以将输入图像与每个核进行卷积，然后取每个像素位置处的最大响应，或者计算梯度的向量和，从而获得完整的边缘图。

## Week2-Image-Enhancement (Monday)

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### Exercise 1 & 2: Contrast and Blur Enhancement

*Enhance the contrast/heavily blurred image using the techniques learned in class, and evaluate the results using a large multimodal model (LMM)...*

使用课堂上学到的技术增强给定图像的对比度/严重模糊的图像，并使用提供质量评分和文本解释的大型多模态模型（LMM）评估结果。

**答案 / Answer: Histogram Equalization & Wiener Deconvolution / 直方图均衡化 (对比度) 与维纳反卷积 (模糊)**

**解析 / Explanation:**

[EN] **Techniques:** For contrast enhancement, techniques like **Histogram Equalization (HE)** or **CLAHE** are optimal. For heavily blurred images, techniques like **Unsharp Masking** or **Wiener Deconvolution** are standard.

[ZH] **技术:** 对于对比度增强，**直方图均衡化 (HE)** 或 **CLAHE** 是最佳选择。对于严重模糊的图像，标准技术包括**非锐化掩蔽 (Unsharp Masking)** 或 **维纳反卷积**。

[EN] **LMM Evaluation:** Input the "Before" and "After" images with a prompt such as: "Rate the visual quality enhancement on a scale of 1-10 and explain the improvements in structural details, noise levels, and contrast."

[ZH] **LMM 评估:** 输入“处理前”和“处理后”的图像，并使用类似提示词：“请在 1-10 分的范围内对视觉质量提升进行评分，并解释在结构细节、噪声水平 and 对比度方面的改善。”

## Open Research Problem: IQA for AI Images

*Imagine you have to design a new IQA metric for modern images generated by AI (e.g., diffusion models). What factors would you include in your metric? How would you ensure it aligns with human subjective evaluation?*

假设你必须为 AI 生成的现代图像设计一种新的图像质量评估 (IQA) 指标。你会在指标中包含哪些因素？如何确保它与人类主观评价保持一致？

**答案 / Answer: Include Perceptual Naturalness and align via RLHF / 包含感知自然度，并通过 RLHF 与人类评价对齐。**

**解析 / Explanation:**

[EN] **Factors:** 1) **Perceptual Naturalness** (lack of AI artifacts like distorted anatomy); 2) **Semantic Alignment** (does the image match the text prompt?); 3) **High-frequency fidelity** (sharpness and texture quality).

[ZH] **包含因素:** 1) **感知自然度** (没有扭曲的解剖结构等 AI 伪影); 2) **语义对齐** (图像是否与文本提示匹配?); 3) **高频保真度** (清晰度和纹理质量)。

[EN] **Human Alignment:** Use **RLHF (Reinforcement Learning from Human Feedback)** or collect a large dataset of human Mean Opinion Scores (MOS) to train a neural network to regress these scores.

[ZH] **人类一致性:** 使用 **RLHF (基于人类反馈的强化学习)**，或者收集大量关于人类平均主观得分 (MOS) 的数据集，训练神经网络来回归这些得分。

这是一份根据你提供的文档内容，按照 LaTeX 模板风格整理的复习问答，涵盖了频率域滤波相关的练习。

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## Frequency Filtering Lab Homework

### 问题 1 / Question 1:

*What does the Fourier Transform do in image processing?* 傅里叶变换在图像处理中有什么作用？

- A. Converts a color image to grayscale (将彩色图像转换为灰度)

- B. Converts an image from the spatial domain to the frequency domain (将图像从空间域转换到频率域)
- C. Enhances image contrast (增强图像对比度)
- D. Removes image noise (去除图像噪声)

答案 / Answer: **B. Converts an image from the spatial domain to the frequency domain**

## 问题 2 / Question 2:

*In the Fourier spectrum of an image, the low-frequency components represent:* 在图像的傅里叶频谱中，低频分量代表什么？

- A. Noise (噪声)
- B. Fine details and edges (精细细节和边缘)
- C. Overall intensity and smooth variations (整体强度和缓慢变化)
- D. Random pixels (随机像素)

答案 / Answer: **C. Overall intensity and smooth variations**

## 问题 3 / Question 3:

*Which statement describes the Convolution Theorem?* 哪项陈述描述了卷积定理？

答案 / Answer: **A. Convolution in spatial domain equals multiplication in frequency domain (空间域的卷积等于频率域的乘法)**

## 问题 4 / Question 4:

*What is the main purpose of a low-pass filter?* 低通滤波器的主要目的是什么？

答案 / Answer: **C. Reduce high-frequency components (减少高频分量)**

## 问题 5 / Question 5:

*Why are notch filters useful in the frequency domain?* 为什么陷波滤波器 (Notch filters)

在频率域中很有用？

**答案 / Answer: B. They remove periodic noise (它们可以去除周期性噪声)**

**解析 / Explanation:**

[EN] Notch filters are designed to block or "notch out" specific frequencies while leaving other frequencies largely unchanged. They are particularly effective at removing **periodic noise** (e.g., interference patterns or repetitive stripes) that manifests as distinct high-energy spikes in the Fourier spectrum. By carefully identifying and masking these specific frequency coordinates, the filter eliminates the noise without significantly affecting the structural integrity of the rest of the image.

[ZH] 陷波滤波器旨在阻断或“切除”特定的频率，同时保留其他频率基本不变。它们在去除**周期性噪声**（例如干扰图案或重复条纹）方面特别有效，这些噪声在傅里叶频谱中表现为明显的能量峰值。通过仔细识别并掩盖这些特定的频率坐标，滤波器能够消除噪声，而不会显著影响图像其余部分的结构完整性。

### 问题 6 / Question 6:

*Why is the Fourier spectrum usually shifted to the center when displayed?* 为什么傅里叶频谱在显示时通常被移动到中心？

**解析 / Explanation:**

[EN] The Fourier spectrum is shifted to the center (using FFT shift) to place the low-frequency (DC) component at the center of the image, making it easier to visualize and apply frequency-domain filters like low-pass or high-pass masks.

[ZH] 傅里叶频谱通常被移位到中心（使用 FFT shift），以便将低频（直流）分量置于图像中心，这样更便于观察以及应用频率域滤波器（如低通或高通掩码）。

### 问题 7 / Question 7:

*Compute the 2D Fourier Transform of a given grayscale image. Display the magnitude spectrum (apply log scaling). Identify where low-frequency and high-frequency components are located.* 计算给定灰度图像的二维傅里叶变换。显示幅度谱（应用对数缩放）。识别低频和高频分量的位置。

**解析 / Explanation:**

[EN] **Magnitude Spectrum Interpretation:** Bright areas at the center of the shifted spectrum represent **low-frequency components** (smooth regions, global intensity). Darker areas towards the edges represent **high-frequency components** (edges, fine





图 1: Enter Caption



图 2: Enter Caption2

details, noise).

[ZH] **幅度谱解读:** 移位后频谱中心的明亮区域代表**低频分量**（平滑区域、全局强度）。向边缘延伸的较暗区域代表**高频分量**（边缘、精细细节、噪声）。

### 问题 8 / Question 8:

*For the given gray image (e.g., cat image), perform frequency domain filtering:* 对于给定的灰度图像（如猫的图片），进行频率域滤波：

1. Compute the Fourier Transform (计算傅里叶变换)
2. Apply a high-pass filter by masking low frequencies (通过掩盖低频应用高通滤波)
3. Perform inverse FFT to obtain the filtered image (执行逆傅里叶变换得到滤波后的图像)

### 解析 / Explanation:

[EN] Analysis of results:



1. **Visible features:** The high-pass filter emphasizes sharp intensity changes, making **edges and fine textures** become more visible. 2. **Flat regions:** Flat regions in the original image have very low frequency content, so they will appear **dark/black** in the high-pass filtered result.

[ZH] 结果分析: 3. 可见特征: 高通滤波器强调了锐利的强度变化, 使**边缘和精细纹理**变得更加明显。4. 平坦区域: 原始图像中的平坦区域具有极低的频率内容, 因此在经过高通滤波的结果中, 这些区域会显示为**暗色/黑色**。由于代表“平坦性”的信息(即低频部分)被滤波器移除或置零, 在进行逆傅里叶变换重构图像时, 这些区域的强度值会趋近于零, 因此在最终生成的图像中就表现为暗色或黑色。本质上, 滤波器“忽略”了背景信息, 只保留了强度剧烈变化的边缘部分。

## Week3-Edge-and-Corner-Detection (Monday)

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### 问题 / Question:

*Can edge detection be performed in frequency domain?*

边缘检测可以在频率域中进行吗?

答案 / Answer: **Yes / 可以**

### 解析 / Explanation:

[EN] Yes, edges correspond to high-frequency components in an image. Edge detection can be done by **high-pass filtering** in the frequency domain. Spatial derivatives (e.g., Sobel, Laplacian) are equivalent to frequency-domain high-pass filters.

[ZH] 可以, 边缘对应于图像中的高频分量。边缘检测可以通过在频率域中进行**高通滤波**来完成。空间导数(如 Sobel、Laplacian)等效于频率域的高通滤波器。

[EN] Steps:

[ZH] 步骤:

[EN] 1. Apply FFT to the image. 2. Multiply by a high-pass filter. 3. Apply inverse FFT to get edge image.

[ZH] 1. 对图像应用快速傅里叶变换 (FFT)。 2. 乘以高通滤波器。 3. 应用逆 FFT 获得边缘图像。

[EN] Limitations:

[ZH] 局限性:

[EN] Less suitable for local edge detection. May introduce ringing artifacts. Spatial intuition is weaker.

[ZH] 不太适合局部边缘检测。可能引入振铃伪影。空间直觉较弱。

## Week3-Local-Features-and-Recognition

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### 问题 / Question:

*Do local features solve the object recognition problem?*

局部特征是否解决了物体识别问题？

答案 / Answer: **Partially / 部分解决**

### 解析 / Explanation:

[EN] Local features (e.g., SIFT, SURF, ORB) **partially solve but do NOT fully solve** the object recognition problem. They provide invariance to translation, rotation, scale, and partial occlusion, but suffer from the semantic gap, intra-class variation, background clutter, and lack of global context.

[ZH] 局部特征（如 SIFT、SURF、ORB）**部分解决但未完全解决**物体识别问题。它们对平移、旋转、尺度和部分遮挡具有不变性，但存在语义鸿沟、类内变化、背景干扰和缺乏全局上下文的问题。

[EN] What they solve well:

[ZH] 已解决的问题:

[EN] Invariance to translation, rotation, and scale. Robustness to partial occlusion and illumination changes. Ability to match local structures across images.

[ZH] 对平移、旋转、尺度变化具有不变性。对部分遮挡和光照变化鲁棒。能够匹配图像间的相同局部结构。

[EN] What they do NOT solve:

[ZH] 未完全解决的问题:

[EN] Semantic gap (low-level vs. high-level semantics). Intra-class variation (different instances of same class). Background clutter (false matches from background). Lack of global context (spatial relationships between parts are lost).

[ZH] 语义鸿沟（低级信息 vs 高级语义）。类内变化（同一类物体的不同实例差异大）。背景干扰（从背景提取的噪声特征）。缺乏全局上下文（丢失了物体部件间的空间关系）。

[EN] **Conclusion:**

[ZH] **结论:**

[EN] Local features are a **critical component** but typically need to be combined with global features, deep learning features, and spatial modeling (e.g., BoW, VLAD, Transformers) for full object recognition.

[ZH] 局部特征是物体识别系统的**重要组成部分**，但通常需要与全局特征、深度学习特征、空间建模（如 BoW、VLAD、Transformer）结合才能解决完整识别问题。

## 问题 / Question:

*How do we deal with the false positives outside the object?*

如何处理物体外部的假阳性匹配？

答案 / Answer: **Geometric verification + clustering / 几何验证 + 聚类**

## 解析 / Explanation:

[EN] False positives outside the object can be handled through multiple strategies. The most effective is **geometric verification** using RANSAC to enforce spatial consistency.

[ZH] 物体外部的假阳性匹配可以通过多种策略处理。最有效的是使用**RANSAC 几何验证**来强制执行空间一致性约束。（相对位置）

[EN] **Methods:**

[ZH] **方法:**

[EN] 1. Geometric verification (RANSAC) to fit transformation models (homography, fundamental matrix).

[ZH] 1. 几何验证（RANSAC）拟合变换模型（单应性矩阵、基本矩阵）。

[EN] 2. Spatial consistency checking (graph matching, spatial verification).

[ZH] 2. 空间一致性检验（图匹配、空间验证）。

[EN] 3. Feature quality filtering (keep high-response DoG extrema, remove edge responses).

[ZH] 3. 特征质量筛选 (保留高响应 DoG 极值点, 去除边缘响应点)。

[EN] 4. Voting and clustering (Hough transform, mean shift, DBSCAN).

[ZH] 4. 投票与聚类 (Hough 变换、均值漂移、DBSCAN)。

[EN] 5. Deep learning approaches (RPN, attention mechanisms).

[ZH] 5. 深度学习方法 (RPN 区域建议网络、注意力机制)。

[EN] **Most effective:** RANSAC + spatial clustering to identify the dominant object region.

[ZH] **最有效方法:** RANSAC + 空间聚类来识别主物体区域。

## 问题 / Question:

*How do we reduce the complexity matching two sets of local features?*

如何降低两组局部特征匹配的复杂度?

答案 / Answer: **Approximate nearest neighbor search / 近似最近邻搜索**

## 解析 / Explanation:

[EN] For two feature sets of size  $n$ , brute-force matching complexity is  $O(n^2)$ . Complexity can be reduced through indexing, quantization, and dimensionality reduction.

[ZH] 对于大小为  $n$  的两组特征集, 暴力匹配复杂度为  $O(n^2)$ 。可以通过索引、量化和降维来降低复杂度。

[EN] **Methods:**

[ZH] **方法:**

[EN] 1. Approximate nearest neighbor search (KD-Tree:  $O(n \log n)$ , FLANN, LSH).

[ZH] 1. 近似最近邻搜索 (KD-Tree:  $O(n \log n)$ 、FLANN、局部敏感哈希 LSH)。

[EN] 2. Feature quantization with inverted index (K-means clustering, Bag of Visual Words, Vocabulary Tree).

[ZH] 2. 特征量化与倒排索引 (K-means 聚类、视觉词袋 BoVW、词汇树)。

[EN] 3. Feature selection & dimensionality reduction (PCA dimensionality reduction,

Top-K strongest features, binary features with Hamming distance).

[ZH] 3. 特征筛选与降维 (PCA 降维、Top-K 最强特征、二值化特征与汉明距离)。

[EN] 4. Hierarchical matching strategies (cascade matching: coarse-to-fine).

[ZH] 4. 分层匹配策略 (级联匹配: 由粗到精)。

[EN] 5. GPU parallel acceleration.

[ZH] 5. GPU 并行加速。

[EN] **Key trade-off:** Accuracy vs. speed. Approximate methods are much faster but may miss the exact nearest neighbor.

[ZH] **关键权衡:** 精度 vs 速度。近似方法快得多但可能错过精确最近邻。

## Week5-Deep Learning Basics-1

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### 第 11 页选择题 1 / Page 11 Quiz 1:

*Which of the following steps is typically REMOVED or BYPASSED in a modern End-to-End Deep Learning pipeline compared to traditional computer vision?*

与传统计算机视觉相比，以下哪一步在现代端到端深度学习流水线中通常被移除或绕过？

- A) Data Collection (数据收集)
- B) Manual Feature Design (e.g., SIFT, HOG) (人工特征设计)
- C) Classifier Training (分类器训练)
- D) Evaluation (评估)

**答案 / Answer: B) Manual Feature Design (e.g., SIFT, HOG) / 人工特征设计**

### 第 11 页选择题 2 / Page 11 Quiz 2:

*Which of the following is the best description of deep learning?*

以下哪项是对深度学习的最佳描述？

- A. Manually designing features and then training a classifier (手动设计特征然后训练分类器)

- B. Learning both feature representations and classifiers from data (从数据中同时学习特征表示和分类器)
- C. Using larger images for classification (使用更大的图像进行分类)
- D. Applying image filters before recognition (在识别之前应用图像滤波器)

**答案 / Answer: B. Learning both feature representations and classifiers from data (从数据中同时学习特征表示和分类器)**

## Shallow vs. Deep —d-bit Parity

*Based on the d-bit parity function example, if we insist on using a shallow structure (only one hidden layer) to represent a highly complex function, what is the theoretical consequence?*

基于 d-bit 奇偶校验函数的例子，如果我们坚持使用浅层结构（仅一个隐藏层）来表示一个高度复杂的函数，理论上会导致什么后果？

- A) The network will train much faster. / 网络训练会快很多
- B) The network requires an exponentially large number of computational nodes (hidden units). / 网络需要指数级数量的计算节点（隐藏单元）
- C) The network will automatically become linear. / 网络会自动变成线性的
- D) The network will completely fail to output any result. / 网络完全无法输出任何结果

**答案 / Answer: B) The network requires an exponentially large number of computational nodes (hidden units). / 网络需要指数级数量的隐藏单元**

## 解析 / Explanation:

[EN] The d-bit parity function is not linearly separable. A shallow network with one hidden layer can represent it in theory, but requires an exponential number of hidden units in  $d$ . Deep networks can represent the same function with  $\mathcal{O}(d)$  units by reusing intermediate features layer by layer.

[ZH] d-bit 奇偶校验函数是线性不可分的。单隐藏层的浅层网络在理论上可以表示它，但需要的隐藏单元数量随  $d$  呈指数增长。深度网络通过逐层重用中间特征，可以用  $\mathcal{O}(d)$  个单元表示同一函数。

## 第 40 页选择题 / Page 40 Quiz:

*According to theoretical analysis in deep learning, which of the following best explains why deep networks can be more efficient than shallow networks for certain tasks?*

根据深度学习的理论分析，以下哪项最能解释为什么深度网络在某些任务中比浅层网络更高效？

- A) Deep networks always have fewer parameters than shallow networks (深度网络的参数总是比浅层网络少)
- B) Deep networks can reuse intermediate feature representations layer by layer, allowing complex functions to be represented with far fewer neurons (深度网络可以逐层重用中间特征表示，从而用更少的神经元表示复杂函数)
- C) Shallow networks cannot use ReLU activation functions (浅层网络不能使用 ReLU 激活函数)
- D) Deep networks do not require training and can perform inference directly (深度网络不需要训练，可以直接进行推理)

**答案 / Answer: B) Deep networks can reuse intermediate feature representations layer by layer... / 深度网络可以逐层重用中间特征表示...**

**解析 / Explanation:**

[EN] Deep networks can reuse intermediate feature representations layer by layer, allowing complex functions to be represented with far fewer neurons.

[ZH] 深度网络可以逐层重用中间特征表示，从而用更少的神经元表示复杂函数。

## 第 57 页 Exercise: Forward Pass Calculation

*Let's calculate the output of a single hidden unit  $j$ .*

*Given: Input vector:  $\mathbf{x} = [2, -1]$ ; Weights for unit  $j$ :  $\mathbf{w}_j = [0.5, 1]$ ; Bias for unit  $j$ :  $w_{j0} = 0.5$ ;*

*Activation function:  $f(\text{net}) = 1$ , if  $\text{net} \geq 0$ ;  $-1$ , if  $\text{net} < 0$*

*What is the output  $y_j$  of this hidden unit?*

计算单个隐藏单元  $j$  的输出。输入向量:  $\mathbf{x} = [2, -1]$ ; 权重:  $\mathbf{w}_j = [0.5, 1]$ ; 偏置:  $w_{j0} = 0.5$ ; 激活函数:  $\text{net} \geq 0$  时为 1, 否则为  $-1$ 。输出  $y_j$  是多少?



- A) 1    B) -1    C) 0.5    D) 0

答案 / Answer: **A) 1**

解析 / Explanation:

$$net = w_{j0} + \sum_{i=1}^n w_{ji}x_i = 0.5 + (2 \times 0.5) + (-1 \times 1) = 0.5 + 1 - 1 = 0.5$$

[EN] Since  $net \geq 0$ , the activation function outputs **1**.

[ZH] 因为  $net \geq 0$ , 所以激活函数输出为 **1**。

## 第 58 页 Exercise: XOR Problem

*Try to manually find weights for a single layer, then for a hidden layer to solve XOR.*

尝试手动找到单层和隐藏层的权重来解决 XOR 问题。答案 / Answer: **Single layer**

| $x_1$ | $x_2$ | XOR |
|-------|-------|-----|
| 0     | 0     | 0   |
| 0     | 1     | 1   |
| 1     | 0     | 1   |
| 1     | 1     | 0   |

表 1: XOR 问题真值表

**fails (not linearly separable), requires ReLU hidden layer. / 单层无法解决 (线性不可分), 需引入 ReLU 隐藏层。**

解析 / Explanation:

[EN] **1. Single layer:** It is mathematically impossible to find a single set of linear weights  $y = w_1x_1 + w_2x_2 + b$  for XOR, because XOR is **not linearly separable**.

[ZH] **1. 单层网络:** 在数学上不可能为 XOR 找到单组线性权重, 因为 XOR **不是线性可分的** (不能用一条直线分开)。

[EN] **2. Hidden layer (ReLU):** By introducing non-linearity, we can solve it:

[ZH] **2. 隐藏层 (ReLU):** 通过引入非线性激活函数, 我们可以解决该问题:

$$h_1 = \text{ReLU}(w_{11} \cdot x_1 + w_{21} \cdot x_2 + b_1) = \text{ReLU}(x_1 - x_2) \quad (w_{11} = 1, w_{21} = -1, b_1 = 0)$$

$$h_2 = \text{ReLU}(w_{12} \cdot x_1 + w_{22} \cdot x_2 + b_2) = \text{ReLU}(x_2 - x_1) \quad (w_{12} = -1, w_{22} = 1, b_2 = 0)$$

$$y = v_1 \cdot h_1 + v_2 \cdot h_2 + b_3 = h_1 + h_2 \quad (v_1 = 1, v_2 = 1, b_3 = 0)$$

$$\text{ReLU}(x) = \max(0, x)$$

## Backpropagation — 输出层梯度计算

In backpropagation, to update the hidden-to-output weight  $W_{kj}$ , we need to compute the partial derivative of the error with respect to the net activation of the output unit.

Suppose:

- The net activation of output unit  $k$  is  $net_k$
- The output is  $y_k$
- The target is  $t_k$
- The loss function is mean squared error:  $E = \frac{1}{2}(t_k - y_k)^2$
- The activation function is the sigmoid function:  $y_k = \sigma(net_k)$

Which of the following correctly represents  $\frac{\partial E}{\partial net_k}$ ?

在反向传播中，为了更新隐藏层到输出层的权重  $W_{kj}$ ，我们需要计算误差对输出单元净激活的偏导数。假设输出单元  $k$  的净激活为  $net_k$ ，输出为  $y_k$ ，目标值为  $t_k$ ，损失函数为均方误差  $E = \frac{1}{2}(t_k - y_k)^2$ ，激活函数为 Sigmoid 函数  $y_k = \sigma(net_k)$ 。以下哪项正确表示了  $\frac{\partial E}{\partial net_k}$ ?

- A)  $(t_k - y_k) \cdot \sigma(net_k)$
- B)  $(y_k - t_k) \cdot \sigma'(net_k)$
- C)  $(t_k - y_k) \cdot \sigma'(net_k)$
- D)  $(y_k - t_k) \cdot \sigma(net_k)$

答案 / Answer: **B)**  $(y_k - t_k) \cdot \sigma'(net_k)$

解析 / Explanation:

[EN] Using the chain rule:

$$\frac{\partial E}{\partial net_k} = \frac{\partial E}{\partial y_k} \cdot \frac{\partial y_k}{\partial net_k}$$

Given  $E = \frac{1}{2}(t_k - y_k)^2$ , we have:

$$\frac{\partial E}{\partial y_k} = \frac{\partial}{\partial y_k} \left[ \frac{1}{2}(t_k - y_k)^2 \right] = -(t_k - y_k) = y_k - t_k$$

And  $\frac{\partial y_k}{\partial \text{net}_k} = \sigma'(\text{net}_k)$ . Therefore:

$$\frac{\partial E}{\partial \text{net}_k} = (y_k - t_k) \cdot \sigma'(\text{net}_k)$$

Option C differs by a sign.

[ZH] 使用链式法则:

$$\frac{\partial E}{\partial \text{net}_k} = \frac{\partial E}{\partial y_k} \cdot \frac{\partial y_k}{\partial \text{net}_k}$$

已知  $E = \frac{1}{2}(t_k - y_k)^2$ , 计算得:

$$\frac{\partial E}{\partial y_k} = \frac{\partial}{\partial y_k} \left[ \frac{1}{2}(t_k - y_k)^2 \right] = -(t_k - y_k) = y_k - t_k$$

且  $\frac{\partial y_k}{\partial \text{net}_k} = \sigma'(\text{net}_k)$ 。因此:

$$\frac{\partial E}{\partial \text{net}_k} = (y_k - t_k) \cdot \sigma'(\text{net}_k)$$

## 优化器选择题 / Optimizer Quiz

*When training deep neural networks, we may encounter oscillations in a canyon-shaped loss surface, leading to slow convergence. Which of the following optimizers introduces an exponential moving average of past gradients to accelerate updates in consistent directions and dampen oscillations?*

在训练深度神经网络时，我们可能会在峡谷形的损失表面上遇到振荡，导致收敛缓慢。以下哪个优化器引入了历史梯度的指数移动平均，以加速一致方向上的更新并抑制振荡？

- A) Standard SGD (标准随机梯度下降)
- B) SGD with Momentum (带动量的 SGD)
- C) Adam (自适应矩估计)
- D) Batch Gradient Descent (批量梯度下降)

**答案 / Answer: B) SGD with Momentum / 带动量的 SGD**

**解析 / Explanation:**

[EN] SGD with Momentum maintains an exponentially decaying moving average of past

gradients. This velocity term accumulates gradients in consistent directions, smoothing out updates and reducing oscillations in narrow canyons. Adam also uses momentum (first moment), but the question specifically highlights the core mechanism of **exponential moving average of past gradients** —historically, momentum SGD is the classic answer for damping oscillations in canyon-shaped loss surfaces.

[ZH] 带动量的 SGD 维护了历史梯度的指数衰减移动平均。这个速度项在一致的方向上累积梯度，使更新更平滑，并减少狭窄峡谷中的振荡。Adam 也使用了动量（一阶矩），但题目特别强调了**历史梯度的指数移动平均**这一核心机制——从历史上看，动量 SGD 是抑制峡谷形损失表面振荡的经典答案。

## Week5-Deep Learning Basics-2

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### 第 25 页讨论题 / Page 25 Discussion

*Should we reduce the number of parameters when overfitting happens?*

当过拟合发生时，我们应该减少参数数量吗？

答案 / Answer: **Not necessarily.** / 不一定。

解析 / Explanation:

[EN] Reducing the number of parameters is one way to combat overfitting, but it is not always necessary or optimal. Alternative methods include: increasing training data, adding regularization (L1/L2), using Dropout, or early stopping. In some cases (e.g., double descent phenomenon), increasing parameters beyond a certain point can actually reduce test error again.

[ZH] 减少参数数量是应对过拟合的一种方法，但并非总是必要或最优的。替代方法包括：增加训练数据、添加正则化（L1/L2）、使用 Dropout 或早停。在某些情况下（例如双下降现象），将参数数量增加到超过某个临界点后，测试误差反而会再次下降。

### 第 25 页《人工智能基础》往年考题 (3) / Page 25 Previous Exam Question (3)

*Model overfitting occurs. Please list at least two solutions. (1 point for identifying overfitting, 4 points total for solutions)*

模型存在过拟合问题。(1 分) 请列出至少两种解决办法。(每列出一种得 2 分，总共可

获得 4 分)

- Solution 1 / 解决办法一
- Solution 2 / 解决办法二
- Solution 3 / 解决办法三 (可选)
- Solution 4 / 解决办法四 (可选)

答案 / Answer:

1. Increase the amount of training data (data augmentation). / 增加训练数据量 (数据增强)
2. Add regularization (L1 or L2 regularization). / 使用正则化 (L1 或 L2 正则化)
3. Use Dropout (randomly drop neurons during training). / 采用 Dropout 技术 (训练时随机丢弃部分神经元)
4. Reduce model complexity (fewer layers or fewer neurons per layer). / 降低模型复杂度 (减少网络层数或每层神经元数量)

解析 / Explanation:

[EN] More training data helps the model learn more representative features. Regularization adds a penalty term on large weights to constrain model complexity. Dropout prevents over-reliance on specific neurons. Reducing model complexity directly limits the capacity of the model to fit noise.

[ZH] 更多的训练数据可以帮助模型学习更具代表性的特征。正则化通过对大权重添加惩罚项来限制模型复杂度。Dropout 防止模型过度依赖某些特定神经元。降低模型复杂度直接限制了模型拟合噪声的能力。

## 第 36 页思考题 / Page 36 Thought Question

*Any continuous function  $f : \mathbb{R}^N \rightarrow \mathbb{R}^M$  can be realized by a network with one hidden layer (given enough hidden neurons). Why "Deep" neural network instead of "Fat" neural network?*

任何连续函数都可以被单隐藏层网络 (给定足够多的隐藏神经元) 表示。为什么我们使用 “深” 网络而不是 “胖” 网络?

**答案 / Answer:** Deep networks are exponentially more parameter-efficient than shallow networks for certain function families. / 对于某些函数族，深度网络在参数效率上具有指数级优势。

**解析 / Explanation:**

[EN] A single hidden layer may require an exponential number of neurons to represent certain functions (e.g., d-bit parity), whereas deep networks can represent the same function with a linear number of parameters by reusing intermediate features layer by layer. Deep networks also encourage hierarchical feature learning.

[ZH] 单隐藏层表示某些函数（如 d-bit 奇偶校验）可能需要指数级数量的神经元，而深度网络通过逐层重用中间特征，可以用线性数量的参数表示相同的函数。此外，深度网络有利于层次化特征学习。

## 第 40 页选择题 / Page 40 Quiz

*According to theoretical analysis in deep learning, which of the following best explains why deep networks can be more efficient than shallow networks for certain tasks?*

根据深度学习的理论分析，以下哪项最能解释为什么深度网络在某些任务中比浅层网络更高效？

- A) Deep networks always have fewer parameters than shallow networks (深度网络的参数总是比浅层网络少)
- B) Deep networks can reuse intermediate feature representations layer by layer, allowing complex functions to be represented with far fewer neurons (深度网络可以逐层重用中间特征表示，从而用更少的神经元表示复杂函数)
- C) Shallow networks cannot use ReLU activation functions (浅层网络不能使用 ReLU 激活函数)
- D) Deep networks do not require training and can perform inference directly (深度网络不需要训练，可以直接进行推理)

**答案 / Answer:** B) Deep networks can reuse intermediate feature representations layer by layer, allowing complex functions to be represented with far fewer neurons. / 深度网络可以逐层重用中间特征表示，从而用更少的神经元表示复杂函数。

### 解析 / Explanation:

[EN] Deep networks can reuse intermediate feature representations layer by layer, allowing complex functions to be represented with far fewer neurons. This exponential efficiency gain is the key theoretical advantage of depth.

[ZH] 深度网络可以逐层重用中间特征表示，从而用更少的神经元表示复杂函数。这种指数级的效率提升是深度结构的核心理论优势。

## 第 41 页编程题 / Page 41 Programming Exercise

*Implement ResNet by numpy and see the performance difference compared to a plain network.*

用 numpy 实现 ResNet，并观察其与普通网络的性能差异。

**答案 / Answer: No single answer; implementation required. / 无单一答案，需实际编程实现。**

### 解析 / Explanation:

[EN] ResNet introduces skip connections that allow gradients to flow directly through the network, mitigating the vanishing gradient problem and enabling the training of very deep networks. Compared to a plain network, ResNet typically achieves lower training and test error for deep architectures.

[ZH] ResNet 引入了跳跃连接，使梯度能够直接流过网络，缓解了梯度消失问题，使得训练非常深的网络成为可能。与普通网络相比，ResNet 在深层架构上通常能达到更低的训练误差和测试误差。

## 第 47 页 FAQ 问题集 / Page 47 FAQ

*Frequently asked questions about network structure design:*

关于网络结构设计的常见问题：

- **Q1: How many layers? How many neurons for each layer?**  
**中文：**应该用多少层？每层应该用多少个神经元？
- **Q2: Can we design the network structure?**  
**中文：**我们可以手动设计网络结构吗？
- **Q3: Can the structure be automatically determined?**



中文：网络结构可以自动确定吗？

答案 / Answer:

1. Trial and error, guided by intuition and empirical rules. / 通过试错法，结合直觉和经验法则。
2. Yes, e.g., Convolutional Neural Networks (CNN) for image data (next lecture). / 可以，例如针对图像数据的卷积神经网络（CNN）（下一讲内容）。
3. Yes, but not widely studied yet (e.g., Neural Architecture Search / NAS). / 可以，但尚未被广泛研究（例如神经架构搜索 / NAS）。

解析 / Explanation:

[EN] Network structure design is still largely empirical. Deep learning practitioners often rely on existing architectures (ResNet, VGG, Transformer) and adjust them based on task requirements. Automated methods like NAS are emerging but computationally expensive.

[ZH] 网络结构设计在很大程度上仍然依赖经验。深度学习实践者通常基于现有架构（ResNet、VGG、Transformer）并根据任务需求进行调整。像 NAS 这样的自动化方法正在兴起，但计算成本较高。

### 第 53 页思考题 / Page 53 Thought Question

*What is the learning target defined on the training data? (Illustrated with:  $2k \rightarrow 1$ , No ink  $\rightarrow 0$ )*

在训练数据上定义的学习目标是什么？（图示： $2k \rightarrow 1$ , No ink  $\rightarrow 0$ ）

答案 / Answer: **The learning target is the desired output (label) for each input sample, such as classifying whether a digit is "2" (output 1) or not (output 0).** / 学习目标是每个输入样本的期望输出（标签），例如分类一个数字是否是“2”（输出 1）或不是（输出 0）。

解析 / Explanation:

[EN] In supervised learning, the training data consists of input-output pairs. The learning target (ground truth label) defines what the network should predict. The loss function measures the discrepancy between the network's prediction and this target.

[ZH] 在监督学习中，训练数据由输入-输出对组成。学习目标（真实标签）定义了网络

应该预测的内容。损失函数用于衡量网络预测与该目标之间的差异。

## Week6-Convolution Neural Network

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### 第 33 页 Exercise: Why Does Convolution Have Fewer Parameters?

Suppose we have a  $100 \times 100$  RGB image (Input volume:  $100 \times 100 \times 3$ ). We want to connect it to a hidden layer with 100 neurons. Q1: How many learnable parameters if MLP? Q2: How many if Convolutional Layer (100 filters,  $5 \times 5$ )?

输入尺寸为  $100 \times 100 \times 3$ ，隐藏层有 100 个神经元。Q1: 如果是 MLP（全连接层）需要多少参数？Q2: 如果是卷积层（100 个  $5 \times 5$  滤波器）需要多少参数？

答案 / Answer: **MLP: 3,000,000 parameters. CNN: 7,500 parameters.**

解析 / Explanation:

[EN] Q1 (MLP): A fully connected layer connects every input pixel to every neuron.

[ZH] Q1 (全连接层): 全连接层将每个输入像素连接到每个神经元。

$$Parameters = 100 \times 100 \times 3 \times 100 = \mathbf{3,000,000}$$

[EN] Q2 (CNN): A convolutional layer shares weights across the spatial dimensions.

[ZH] Q2 (卷积层): 卷积层在空间维度上共享权重。

$$Parameters = (\text{Filter Size: } 5 \times 5 \times 3) \times (\text{Num Filters: } 100) = \mathbf{7,500}$$

### 第 37 页 Exercise: Output Feature Map Size

What is the output feature map size? (Input:  $7 \times 7$ , Filter:  $3 \times 3$ )

输出特征图尺寸是多少？（输入:  $7 \times 7$ , 滤波器:  $3 \times 3$ ）

Formula / 公式:  $\text{Output\_Size} = \lfloor \frac{W-K+2P}{S} \rfloor + 1$

答案 / Answer: **1)  $5 \times 5$    2)  $9 \times 9$    3)  $5 \times 5$**

解析 / Explanation:

1. [EN] Stride 1, Padding 0:  $\frac{7-3}{1} + 1 = \mathbf{5 \times 5}$   
[ZH] 步长 1, 填充 0: 输出尺寸为  $5 \times 5$
2. [EN] Stride 1, Padding 2:  $\frac{7+2(2)-3}{1} + 1 = \mathbf{9 \times 9}$   
[ZH] 步长 1, 填充 2: 输出尺寸为  $9 \times 9$
3. [EN] Stride 2, Padding 2:  $\lfloor \frac{7+4-3}{2} \rfloor + 1 = \lfloor \frac{8}{2} \rfloor + 1 = \mathbf{5 \times 5}$   
[ZH] 步长 2, 填充 2: 向下取整后输出尺寸为  $5 \times 5$

### 第 43 页 Exercise: Max Pooling (True/False)

*Evaluate the following statements regarding a Max Pooling layer (True / False):*

判断以下关于最大池化层的陈述是真还是假:

1. *After Max Pooling, the number of channels in the feature map will be halved.* (最大池化后, 特征图的通道数会减半。)

答案 / Answer: **False / 错误**

[EN] Max Pooling operates independently on each channel; the number of channels stays the same.

[ZH] Max Pooling 在每个通道上独立操作, 特征图的通道数保持不变。

2. *A Max Pooling layer has NO learnable parameters to update.* (最大池化层没有可学习的参数需要更新。)

答案 / Answer: **True / 正确**

[EN] Max Pooling just takes the maximum value in a window; there are no weights to learn during back-propagation.

[ZH] Max Pooling 仅取窗口内的最大值, 反向传播时没有需要学习的权重。

### 第 47 页 Exercise: CNN Component Order

*Put the following components in the correct order for a typical image classification CNN:*

将以下组件按典型图像分类 CNN 的正确顺序排列: (1) Fully connected layer (全连接层)

(2) Convolutional layer (卷积层)

- (3) Output class probabilities (输出概率)
- (4) Pooling layer (池化层)
- (5) Flatten (展平)
- (6) Input image (输入图像)

**答案 / Answer:** (6) → (2) → (4) → (5) → (1) → (3)

**解析 / Explanation:**

[EN] Input image → Convolutional layer → Pooling layer → Flatten → Fully connected layer → Output class probabilities

[ZH] 输入图像 → 卷积层 → 池化层 → 展平 → 全连接层 → 输出类别概率

## 第 68 页 Exercise: Why stack $3 \times 3$ filters?

*In VGGNet, they used two consecutive  $3 \times 3$  layers to replace one  $5 \times 5$  layer. Do they have the same Receptive Field? Which requires fewer parameters?*

在 VGGNet 中，使用两个连续的  $3 \times 3$  层替换一个  $5 \times 5$  层。它们的感受野相同吗？哪种需要的参数更少？

**答案 / Answer:** Same receptive field, but two  $3 \times 3$  layers require fewer parameters. / 感受野相同，但两个  $3 \times 3$  层参数更少。

**解析 / Explanation:**

[EN] Q1: Same Receptive Field? Yes. Two stacked  $3 \times 3$  layers cover the same spatial area as one  $5 \times 5$  layer.

[ZH] Q1: 感受野相同吗？是的。两个堆叠的  $3 \times 3$  卷积层覆盖的空间区域与一个  $5 \times 5$  层相同。

[EN] Q2: Fewer parameters?

Two  $3 \times 3$  layers:  $2 \times (3 \times 3 \times C \times C) = 18C^2$ .

One  $5 \times 5$  layer:  $5 \times 5 \times C \times C = 25C^2$ .

Therefore, **two  $3 \times 3$  layers require fewer parameters.**

[ZH] Q2: 参数更少吗？

两个  $3 \times 3$  层:  $2 \times (3 \times 3 \times C \times C) = 18C^2$ 。

一个  $5 \times 5$  层:  $5 \times 5 \times C \times C = 25C^2$ 。

因此，**两个  $3 \times 3$  层需要的参数更少。**

## 第 78 页选择题 / Page 78 Quiz: Skip Connections

*The skip connection  $H(x) = F(x) + x$  was introduced primarily to solve which of the following issues?*

跳跃连接  $H(x) = F(x) + x$  的引入主要是为了解决以下哪个问题？

- A. To reduce the number of parameters in the network. (减少网络中的参数数量)
- B. To alleviate the Vanishing Gradient problem in deep networks. (缓解深度网络中的梯度消失问题)
- C. To reduce GPU memory consumption. (减少 GPU 内存消耗)

**答案 / Answer: B. To alleviate the Vanishing Gradient problem in deep networks. (缓解深度网络中的梯度消失问题)**

### 解析 / Explanation:

[EN] By adding the input  $x$  directly to the output of the block  $F(x)$ , gradients can flow directly backwards during training, mitigating the vanishing gradient problem in very deep networks.

[ZH] 通过将输入  $x$  直接加到块的输出  $F(x)$  上，在训练期间梯度可以直接向后流动，从而缓解了极深网络中的梯度消失问题。

## Week7-Vision Transformer

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### 第 16 页选择题 / Page 16 Quiz:

*Which of the following statements about the Transformer is correct?*

关于 Transformer，以下哪项陈述是正确的？

- A. The Transformer only contains an encoder. (Transformer 仅包含编码器。)
- B. The Transformer uses recurrence to process tokens one by one. (Transformer 使用循环方式逐个处理 token。)
- C. The encoder and decoder are both composed of stacked blocks. (编码器和解码器均由堆叠的块组成。)

器都由堆叠的块组成。)

- D. Positional information is unnecessary in Transformer. (在 Transformer 中位置信息是不必要的。)

**答案 / Answer: C. The encoder and decoder are both composed of stacked blocks. (编码器和解码器都由堆叠的块组成。)**

## 第 27 页选择题 / Page 27 Quiz: Why multi-head attention?

*What is the main benefit of multi-head attention?*

多头注意力的主要好处是什么?

- A. It reduces the input sequence length. (它减少了输入序列的长度。)
- B. It allows the model to focus on different relationships in parallel. (它允许模型并行关注不同的关系。)
- C. It removes the need for positional encoding. (它消除了对位置编码的需求。)
- D. It makes the model equivalent to CNN. (它使模型等同于 CNN。)

**答案 / Answer: B. It allows the model to focus on different relationships in parallel. (它允许模型并行关注不同的关系。)**

## 第 34 页选择题 / Page 34 Quiz: Positional encoding

*Why does a Transformer need positional encoding?*

为什么 Transformer 需要位置编码?

- A. Because token embeddings are always zero. (因为 token 嵌入总是零。)
- B. Because self-attention alone does not explicitly encode word order. (因为自注意力本身不显式编码词序。)
- C. Because positional encoding reduces the number of parameters. (因为位置编码减少了参数数量。)
- D. Because positional encoding replaces the feed-forward network. (因为位置编码替换了前馈网络。)

**答案 / Answer: B. Because self-attention alone does not explicitly encode**

word order. (因为自注意力本身不显式编码词序。)

## 第 44 页选择题 / Page 44 Quiz: Layer Normalization

*Which statement is the best description of Layer Normalization?*

哪项陈述是对层归一化的最佳描述?

- A. It normalizes features within a sample. (它在一个样本内归一化特征。)
- B. It normalizes all training images in the dataset. (它归一化数据集中的所有训练图像。)
- C. It is only used in CNNs. (它仅用于 CNN。)
- D. It changes the input sequence length. (它改变输入序列长度。)

答案 / Answer: **A. It normalizes features within a sample. (它在一个样本内归一化特征。)**

## 第 55 页 Exercise: Patch calculation in ViT (Python)

*Write a Python function that takes an image of shape  $(H, W, C)$  and a patch size  $P$ , and returns all non-overlapping patches.*

编写一个 Python 函数，输入形状为  $(H, W, C)$  的图像和 patch 大小  $P$ ，返回所有不重叠的图像块 (Patches)。

### Python Solution

```
import numpy as np

def extract_patches(image, P):
    # EN: image shape is (H, W, C)
    # ZH: 图像形状为 (高度, 宽度, 通道数)
    H, W, C = image.shape
    patches = []

    # EN: Iterate through the image with step size P (non-overlapping)
    # ZH: 以步长P遍历图像 (确保无重叠)
    for i in range(0, H, P):
        for j in range(0, W, P):
            patch = image[i:i+P, j:j+P, :]
            patches.append(patch)

    return np.array(patches) # Output shape: (N, P, P, C)
```

Why relative positional encoding? 为什么需要相对位置编码? To equip self-attention with an important property of CNN, i.e. translation equivariance. 为自注意力赋予 CNN 的一个重要特性, 即平移等变性。Translation equivariance: If the input image is translated, the output feature map will also be translated by the same amount. Translation invariance: the model's recognition result (e.g., classification label) does not change when the input is translated. 平移等变性: 如果输入图像被平移, 输出的特征图也会被相应地平移。平移不变性: 当输入被平移时, 模型的识别结果 (如分类标签) 不会改变。

### 第 65 页选择题 / Page 65 Quiz:

*Which statement is generally true?*

哪项陈述总体上是正确的?

- A. CNN is better at modeling long-range dependency than Transformer. (CNN 比 Transformer 更擅长建模长距离依赖。)
- B. Transformer naturally focuses on local neighborhoods only. (Transformer 自然地仅关注局部邻域。)
- C. CNN is usually more computation-efficient than vanilla Transformer. (CNN 通常比原始 Transformer 计算效率更高。)



- D. Transformer does not need positional information in vision tasks. (Transformer 在视觉任务中不需要位置信息。)

答案 / Answer: **C. CNN is usually more computation-efficient than vanilla Transformer.** (CNN 通常比原始 Transformer 计算效率更高。)

## Week8-Object Detection

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### 第 4 页: Classification vs Detection

*What makes object detection harder than image classification?*

为什么目标检测比图像分类更难?

答案 / Answer: **Detection outputs both category and bounding box.** / 检测需同时输出类别和位置框 (因此更难)。

解析 / Explanation:

[EN] 1. Image classification outputs only the object category.

2. Object detection outputs both the category and the bounding box location.

3. It is harder because the model must simultaneously determine *what* the object is and *where* it is located.

[ZH] 1. 图像分类仅输出物体类别。

2. 目标检测必须同时输出每个物体的类别和边界框位置。

3. 目标检测更难，因为模型必须同时确定物体“是什么”以及它“在哪里”。

### 第 8 页: Region Proposals

Suppose we use exhaustive sliding-window search on an image. 假设我们在图像上使用穷举滑动窗口搜索。 Why is this method computationally expensive? 为什么这种方法计算代价高? Why do we need windows of different sizes? 为什么我们需要不同尺寸的窗口? How do region proposals reduce the search cost? 区域提议如何降低搜索成本?

答案 Exhaustive search is expensive because the detector must evaluate a very large number of candidate windows. 穷举搜索代价高是因为检测器必须评估大量的候选窗口。

Different window sizes are needed because objects may appear at different scales. 需要不同窗口尺寸是因为物体可能以不同尺度出现。 Region proposals reduce the search

cost by selecting a smaller set of likely object regions before classification. 区域提议通过在分类前选择一小部分较可能是物体的区域来降低搜索成本。

*Why do we need region proposals?*

为什么我们需要区域提议？

**答案 / Answer: Reduce computation cost of exhaustive search. / 降低穷举搜索的计算成本。**

**解析 / Explanation:**

[EN] Exhaustive sliding-window search requires evaluating an impractically large number of candidate windows.

Different window sizes are needed because objects appear at different scales.

Region proposals significantly reduce the search cost by filtering out background and selecting a smaller set of highly likely object regions before running the heavy classification network.

[ZH] 穷举滑动窗口搜索需要评估数量极其庞大的候选窗口，计算代价过高。

需要不同窗口尺寸是因为图像中的物体可能以不同的尺度出现。

区域提议通过过滤掉背景，并在运行繁重的分类网络之前选择一小部分极可能是物体的区域，从而显著降低了搜索成本。

## 第 14 页: Selective Search Pipeline Order

*Reorder the following steps for Selective Search:*

为选择性搜索 (Selective Search) 重新排列以下步骤：

- A. Merge the most similar neighboring regions (合并最相似的相邻区域)
- B. Generate initial over-segmentation (生成初始过分割)
- C. Collect region proposals (收集区域提议)
- D. Repeat the merging process (重复合并过程)

**答案 / Answer: B → A → D → C**

**解析 / Explanation:**

[EN] Generate initial over-segmentation → Merge the most similar neighboring regions → Repeat the merging process → Collect region proposals.

[ZH] 生成初始过分割 → 合并最相似的相邻区域 → 重复合并过程 → 收集区域提议。

## 第 18 页: Two-stage vs One-stage

*What is the key idea of two-stage and one-stage detectors? Is R-CNN two-stage or one-stage?*

两阶段和单阶段检测器的核心思想是什么? R-CNN 属于哪种?

**答案 / Answer: R-CNN is a Two-stage detector. / R-CNN 是两阶段检测器。**

**解析 / Explanation:**

[EN] **Two-stage:** Generates category-independent region proposals first, and then classifies them.

**One-stage:** Skips the proposal stage entirely and predicts bounding boxes directly on dense feature map locations.

**R-CNN:** is a Two-stage detector because it uses Selective Search to generate proposals before feeding them to a CNN.

[ZH] **两阶段 (Two-stage):** 首先生成与类别无关的区域提议, 然后再对这些提议进行分类。

**单阶段 (One-stage):** 完全跳过提议阶段, 直接在密集的特征图位置上预测边界框。

**R-CNN:** 属于两阶段检测器, 因为它在将图像输入 CNN 之前, 先使用选择性搜索 (Selective Search) 来生成提议。

## Week9-Practical Issues

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### 第 17 页: Imbalance Ratio

*A medical dataset contains four classes: Normal(5000), Mild Disease(1200), Severe Disease(300), Rare Disease(50). Compute Imbalance Ratio.*

一个医学数据集包含四类: 正常 (5000)、轻度疾病 (1200)、重度疾病 (300)、罕见疾病 (50)。计算不平衡比率 (Imbalance Ratio)。

**答案 / Answer: Majority: Normal(5000), Minority: Rare(50),  $IR = 100$ . / 多数类: 正常 (5000), 少数类: 罕见 (50), 不平衡比率  $IR = 100$ 。**

**解析 / Explanation:**

[EN] The imbalance ratio is defined as the number of samples in the majority class divided by the number of samples in the minority class:  $IR = \frac{5000}{50} = 100$ . Yes, this dataset is severely imbalanced.

[ZH] 不平衡比率定义为多数类的样本数除以少数类的样本数:  $IR = \frac{5000}{50} = 100$ 。是的, 这个数据集存在严重的不平衡。

## 第 33 页: Feature Extractor or Classifier?

*Decide whether it mainly affects the feature extractor or classifier:*

判断它主要影响特征提取器还是分类器:

- ImageNet-pretrained backbone (使用 ImageNet 预训练骨干网络) → 答案 / Answer: **Feature extractor (特征提取器)**
- Freezing backbone, retraining FC layer (冻结骨干网络, 重训全连接层) → 答案 / Answer: **Classifier (分类器)**
- Data augmentation (数据增强) → 答案 / Answer: **Feature extractor (特征提取器)**
- Replacing final head (替换最终分类头) → 答案 / Answer: **Classifier (分类器)**

## 第 42 页: Source vs Target Domain

*Identify the source domain and target domain:*

识别源域 (Source) 和目标域 (Target):

- Train on daytime, deploy on nighttime (白天训练, 夜间部署) → 答案 / Answer: **Source: daytime (白天); Target: nighttime (夜间)**
- Synthetic game to real urban (合成游戏到真实城市) → 答案 / Answer: **Source: synthetic game (合成游戏); Target: real urban (真实城市)**
- MNIST to SVHN (从 MNIST 到 SVHN) → 答案 / Answer: **Source: MNIST;**

**Target: SVHN**

- Western face to Asian face (西方人脸到亚洲人脸) → **答案 / Answer: Source: Western face (西方人脸); Target: Asian face (亚洲人脸)**

### 第 63 页选择题 / Page 63 Quiz: Pixel-space alignment

*Which statement best describes pixel-space alignment?*

哪项陈述最能描述像素空间对齐?

- A. It aligns intermediate feature distributions only. (它仅对齐中间特征分布。)
- B. It changes the visual style of images to reduce domain gap. (它改变图像的视觉风格以减少域差异。)
- C. It removes minority classes from the dataset. (它从数据集中移除少数类。)
- D. It only changes the final classification layer. (它仅改变最后的分类层。)

**答案 / Answer: B. It changes the visual style of images to reduce domain gap. (改变图像的视觉风格以减少域差异。)**

### 第 68 页选择题 / Page 68 Quiz: Why is category shift challenging?

*Why is category shift challenging in domain adaptation?*

为什么类别偏移在域自适应中具有挑战性?

- A. The source and target domains always have exactly the same classes. (源域和目标域总是具有完全相同的类别。)
- B. The target domain may contain unknown classes not present in the source domain. (目标域可能包含源域中不存在的未知类别。)
- C. The model does not need any feature extractor. (模型不需要任何特征提取器。)
- D. The training set is always balanced. (训练集总是平衡的。)

**答案 / Answer: B. The target domain may contain unknown classes not present in the source domain. (目标域可能包含源域中不存在的未知类别。)**

## Week10-Practical-Issues (Thursday)

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### 第 44 页 Q1: Principle of learning from unlabeled data?

*Across all methods (MIL, consistency regularization, self-training), what is the common principle that allows learning from unlabeled or weakly labeled data?*

在所有方法中，允许从未标记或弱标记数据中学习的共同原理是什么？

答案 / Answer: **Impose structural constraints / 对预测空间施加结构约束**

解析 / Explanation:

[EN] We do not rely on explicit labels, but instead impose structural constraints on the prediction space to extract learning signals directly from the data distribution itself.

[ZH] 我们不依赖于显式的标签，而是通过对预测空间施加结构约束，直接从数据分布本身中提取学习信号。

### 第 46 页 Q2: Better models vs better supervision?

*Which is more fundamental in weak/semi-supervised learning: better models or better supervision design? Why?*

在弱/半监督学习中，哪个更根本：更好的模型还是更好的监督设计？为什么？

答案 / Answer: **Better supervision defines the target. / 监督设计更根本，因为它定义了学习目标。**

解析 / Explanation:

[EN] Better models improve optimization efficiency and representational capacity, but better supervision is more fundamental because it defines the learning target itself.

[ZH] 更好的模型提高了优化效率和表示能力，但更好的监督设计更为根本，因为它定义了学习目标本身。

### 第 48 页 Q3: System design with unlimited compute but limited labels?

*If you had unlimited computation but very limited labeled data, how would you design a vision learning system today?*

如果你拥有无限的计算资源但非常有限的标注数据，你今天会如何设计一个视觉学习系统？

**答案 / Answer: Self-Supervised Learning (SSL) paradigm / 使用自监督学习范式构建基础模型。**

**解析 / Explanation:**

[EN] I would use a **Self-Supervised Learning (SSL)** paradigm to build a Foundation Model.

1. **Pre-training:** Use massive amounts of unlabeled data and compute to train a large Vision Transformer using Masked Image Modeling (e.g., MAE) or Contrastive Learning.
2. **Fine-tuning:** Use the limited labeled data for Parameter-Efficient Fine-Tuning (PEFT).
3. **Self-Training:** Use the fine-tuned model to generate pseudo-labels on unlabeled data, and iteratively retrain.

[ZH] 我会使用自监督学习 (SSL) 范式来构建一个基础模型。

1. **预训练:** 利用海量无标签数据和强大算力，通过掩码图像建模（如 MAE）或对比学习训练大型 Vision Transformer。
2. **微调:** 仅使用有限的有标签数据进行参数高效微调 (PEFT)。
3. **自训练:** 利用微调后的模型在无标签数据上生成伪标签，并进行迭代训练以进一步提升性能。

## Week11-Practical Issues (Self-Supervised Learning)

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### 第 7 页: 判断自监督任务 / Exercise: Identify Self-Supervised Tasks

*Determine whether each of the following tasks belongs to self-supervised learning, and explain why.*

判断以下每个任务是否属于自监督学习，并解释原因。

- 给定手动标注为“猫”或“狗”的图片集，训练一个分类器。→ 答案 / Answer: **No (否)**
- 随机将图片旋转  $0^\circ$ ,  $90^\circ$ ,  $180^\circ$ , 或  $270^\circ$ ，训练模型预测旋转角度。→ 答案 / Answer: **Yes (是)**
- 将彩色图转为灰度图，训练模型恢复原始颜色。→ 答案 / Answer: **Yes (是)**
- 使用 ImageNet 类别标签训练 ResNet 分类器。→ 答案 / Answer: **No (否)**
- 随机遮挡图像的一部分，训练模型重建被遮挡区域。→ 答案 / Answer: **Yes (是)**

**解析 / Explanation:**

[EN] Self-supervised learning generates labels automatically from the data itself. Rotation prediction, colorization, and inpainting use intrinsic data properties to create supervision. Manual labels (cat/dog, ImageNet) belong to supervised learning.

[ZH] 自监督学习从数据本身自动生成标签。旋转预测、着色和修复利用数据的内在属性来创建监督信号。手动标签（猫/狗、ImageNet）属于监督学习。

## 第 24 页: MAE 编码器输入 / MAE Encoder Input

*In MAE, which patches are fed into the encoder during pre-training?*

在 MAE 预训练期间，哪些图像块被输入到编码器中？

- A. All patches, including masked patches (所有块，包括被遮盖的块)
- B. Only the visible patches (仅可见块)
- C. Only the masked patches (仅被遮盖的块)
- D. Random noise patches (随机噪声块)

答案 / Answer: **B. Only the visible patches (仅可见块)**



### 解析 / Explanation:

[EN] MAE uses an asymmetric encoder-decoder architecture. The encoder processes only the visible patches, while the decoder, which is much smaller, handles the full set including mask tokens.

[ZH] MAE 使用非对称的编码器-解码器架构。编码器只处理可见的图像块，而小得多的解码器则处理包括掩码标记在内的全部集合。

## 第 30 页: MAE 掩码比例分析 / MAE Mask Ratio Analysis

*Answer the following questions about mask ratio in MAE:*

回答关于 MAE 中掩码比例的问题：

1. Why might performance be unsatisfactory when the mask ratio is too low?  
为什么当掩码比例太低时性能可能不理想？
2. Why might performance also drop when the mask ratio is too high?  
为什么当掩码比例太高时性能也会下降？
3. According to the results, what is a reasonable recommended mask ratio for MAE?  
根据结果，MAE 的合理推荐掩码比例是多少？

### 答案 / Answer:

1. If the mask ratio is too low, the reconstruction task may be too easy, so the model may not learn strong semantic representations. / 如果掩码比例过低，重建任务可能过于简单，模型可能无法学习到强大的语义表征。
2. If the mask ratio is too high, too little visible information remains, making the reconstruction task overly difficult. / 如果掩码比例过高，剩余的可见信息太少，使得重建任务过于困难。
3. Around 75% is generally a reasonable choice. / 通常 75% 左右是合理的选择。

## 第 55 页: MoCo 动量编码器的作用 / Purpose of MoCo Momentum Encoder

*In MoCo, why is a momentum encoder used?*

在 MoCo 中，为什么使用动量编码器？

- A. To make the model smaller (使模型更小)
- B. To generate more image augmentations (生成更多图像增强)
- C. To maintain consistent representations for the queue (为队列保持一致的表示)
- D. To reconstruct masked patches (重建被遮盖的块)

**答案 / Answer: C. To maintain consistent representations for the queue (为队列保持一致的表示)**

**解析 / Explanation:**

[EN] The momentum encoder is updated slowly using a moving average of the query encoder's weights ( $\theta_k \leftarrow m\theta_k + (1 - m)\theta_q$ ). This provides stable and consistent representations for the keys in the queue, which is crucial for effective contrastive learning.

[ZH] 动量编码器使用查询编码器权重的移动平均值进行缓慢更新，这为队列中的键(keys) 提供了稳定且一致的表示，这对于有效的对比学习至关重要。

## 第 59 页: 对比学习方法分类 / Categorizing Contrastive Methods

Complete the table: Does each method use negative examples? Does it use a momentum encoder?

填写表格：每种方法是否使用负例？是否使用动量编码器？

| Method  | Uses negative examples? | Uses momentum encoder? |
|---------|-------------------------|------------------------|
| SimCLR  | 答案 / Answer: Yes (是)    |                        |
|         | 答案 / Answer: No (否)     |                        |
| MoCo    | 答案 / Answer: Yes (是)    |                        |
|         | 答案 / Answer: Yes (是)    |                        |
| BYOL    | 答案 / Answer: No (否)     |                        |
|         | 答案 / Answer: Yes (是)    |                        |
| SimSiam | 答案 / Answer: No (否)     |                        |
|         | 答案 / Answer: No (否)     |                        |

解析 / Explanation:

[EN] SimCLR uses in-batch negatives but no momentum encoder. MoCo uses a queue of negatives with a momentum encoder. BYOL removes negative examples but keeps the momentum encoder. SimSiam further simplifies by removing both.

[ZH] SimCLR 使用批次内负例但没有动量编码器。MoCo 使用带有动量编码器的负例队列。BYOL 移除了负例但保留了动量编码器。SimSiam 通过移除两者进一步简化。

## Week11-Thursday-Lightweight Computer Vision

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### 第 32 页: 延迟的定义 / Definition of Latency

*How is Latency defined in the context of computational speed?*

在计算速度的上下文中，延迟是如何定义的？

答案 / Answer: **Latency** =  $\frac{\text{FLOPs}}{\text{FLOPS}}$

解析 / Explanation:

[EN] Latency is defined as the ratio of the total number of operations (FLOPs) to the execution speed (FLOPS). Low FLOPs or low FLOPS (computational speed) can both lead to high latency.

[ZH] 延迟定义为总操作数 (FLOPs) 与执行速度 (FLOPS) 的比值。低 FLOPs 或低 FLOPS (计算速度) 都可能导致高延迟。

### 第 34 页: 标准卷积 vs 深度卷积 / Standard Conv vs Depthwise Conv

*Given an input  $I \in \mathbb{R}^{c \times h \times w}$  and a kernel size  $k$ , compare the FLOPs of a standard convolution and a depthwise convolution.*

给定输入  $I \in \mathbb{R}^{c \times h \times w}$  和卷积核大小  $k$ ，比较多标准卷积和深度卷积的 FLOPs。

答案 / Answer:

- **Standard Conv FLOPs:**  $h \times w \times k^2 \times c^2$  (标准卷积)
- **Depthwise Conv FLOPs:**  $h \times w \times k^2 \times c$  (深度卷积)

解析 / Explanation:

[EN] Standard convolution applies  $c$  filters, each operating on all  $c$  input channels, resulting in a quadratic cost ( $c^2$ ). Depthwise convolution applies a single filter per input channel, leading to a linear cost ( $c$ ).

[ZH] 标准卷积应用  $c$  个滤波器，每个滤波器对所有  $c$  个输入通道进行操作，导致二次成本。深度卷积对每个输入通道应用一个独立的滤波器，产生线性成本。

## 第 39 页: 部分卷积 / Partial Convolution (PConv)

*How does Partial Convolution (PConv) reduce computational cost compared to regular convolution?*

与常规卷积相比，部分卷积如何降低计算成本？

**答案 / Answer: PConv applies regular convolution on only a part of the input channels (e.g.,  $c_p$  channels) for spatial feature extraction, leaving the remaining channels untouched. / PConv 仅对部分输入通道应用常规卷积以提取空间特征，保持其余通道不变。**

**解析 / Explanation:**

[EN] By processing only  $c_p$  representative channels instead of all  $c$  channels, PConv significantly reduces FLOPs to  $h \times w \times k^2 \times c_p^2$  and memory access, achieving lower latency. A subsequent pointwise convolution then fuses information across all channels.

[ZH] 通过仅处理  $c_p$  个代表性通道而非所有  $c$  个通道，PConv 显著降低了 FLOPs 至  $h \times w \times k^2 \times c_p^2$  和内存访问，实现了更低的延迟。随后的逐点卷积再跨所有通道融合信息。

## 第 51 页: Lite DETR 效果 / Lite DETR Performance

*What is the main benefit of Lite DETR?*

Lite DETR 的主要优点是什么？

**答案 / Answer: Lite DETR can effectively reduce the GFLOPs of the detection head by 60% while keeping 99% of the original performance. / Lite DETR 能够将检测头的 GFLOPs 有效降低 60%，同时保持 99% 的原始性能。**

**解析 / Explanation:**

[EN] Lite DETR proposes an interleaved update scheme, updating high-level features more frequently than low-level features. This significantly reduces the number of feature

tokens processed, leading to high efficiency without sacrificing much accuracy.

[ZH] Lite DETR 提出了一种交错更新方案，高频更新高级特征，低频更新低级特征。这显著减少了需要处理的特征标记数量，从而在不牺牲太多精度的情况下实现了高效率。

## 第 55 页: Lite DETR 动机 / Motivation for Lite DETR

*What limitation of standard DETR does Lite DETR address?*

Lite DETR 解决了标准 DETR 的哪个局限性？

答案 / Answer: **Standard DETR only uses high-level features with low resolution, lacking rich local details critical for small object detection. Lite DETR incorporates multi-scale features efficiently.** / 标准 DETR 仅使用低分辨率的高级特征，缺少对小目标检测至关重要的丰富局部细节。Lite DETR 高效地融合了多尺度特征。

解析 / Explanation:

[EN] DETR uses a Transformer on a single-scale, low-resolution feature map, which is poor for small objects. Lite DETR designs an efficient encoder block that interleaves updates of high-level and low-level features, capturing both global context and local details.

[ZH] DETR 在单尺度、低分辨率的特征图上使用 Transformer，这对小目标效果不佳。Lite DETR 设计了一种高效的编码器模块，交错更新高级和低级特征，从而同时捕捉全局上下文和局部细节。

## Week12-Generative Computer Vision (Unconditional Generative Models)

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### 第 12 页: 生成模型的目标 / Goal of Generative Model

*Given a dataset containing many dog images, what is the main goal of a generative model?*

给定一个包含许多狗图像的数据集，生成模型的主要目标是什么？

- A. To memorize all training images and copy them during testing (记住所有训练图像并在测试时复制它们)
- B. To learn the underlying data distribution and generate new samples from it (学

习底层数据分布并从中生成新样本)

- C. To only learn low-level features such as edges and textures (仅学习边缘和纹理等低级特征)
- D. To predict a class label for each input image (为每个输入图像预测类别标签)

答案 / Answer: **B**

解析 / Explanation:

[EN] The fundamental goal of a generative model is to estimate the true data distribution  $p_{\text{data}}(x)$  from training samples, enabling the generation of new, unseen samples that resemble the training data.

[ZH] 生成模型的基本目标是从训练样本中估计真实的数据分布  $p_{\text{data}}(x)$ ，从而能够生成与训练数据相似的全新、未见过的样本。

## 第 18 页: 自编码器为何不足以用于生成? / Why is Autoencoder not enough for generation?

*An Autoencoder can encode an image into a latent vector and decode it back into an image. Why is it still not necessarily a good generative model?*

自编码器可以将图像编码为潜在向量并解码回图像。为什么它仍然不一定是一个好的生成模型?

- A. Because Autoencoders do not have an encoder (因为自编码器没有编码器)
- B. Because Autoencoders mainly optimize reconstruction error and do not guarantee a continuous, well-organized latent space (因为自编码器主要优化重建误差, 不保证连续、良好组织的潜在空间)
- C. Because Autoencoders cannot process image data (因为自编码器无法处理图像数据)
- D. Because the decoder cannot generate images (因为解码器无法生成图像)

答案 / Answer: **B**

解析 / Explanation:

[EN] Autoencoders focus solely on minimizing reconstruction loss without regularizing the latent space structure. Consequently, arbitrary points sampled from the latent space may decode to meaningless or unrealistic outputs.

[ZH] 自编码器仅专注于最小化重建损失，而不对潜在空间结构进行正则化。因此，从潜在空间中任意采样的点可能会解码出无意义或不真实的输出。

## 第 23 页: VAE 中 KL 散度的作用 / Role of KL Divergence in VAE

*In a VAE, the loss function is often written as  $\mathcal{L} = \text{Reconstruction Loss} + D_{KL}(q(z|x)||p(z))$ .*

*What is the main role of the KL divergence term?*

在 VAE 中，损失函数通常写为  $\mathcal{L} = \text{Reconstruction Loss} + D_{KL}(q(z|x)||p(z))$ 。KL 散度项的主要作用是什么？

- A. To make the decoder deeper (使解码器更深)
- B. To make the encoder's latent distribution close to a prior distribution, usually a standard Gaussian (使编码器的潜在分布接近先验分布，通常是标准高斯分布)
- C. To guarantee that generated images are always sharper (保证生成的图像总是更清晰)
- D. To remove the need for training data (消除对训练数据的需求)

答案 / Answer: **B**

解析 / Explanation:

[EN] The KL divergence regularizes the learned latent distribution  $q(z|x)$  to be close to a standard Gaussian prior  $p(z) \sim \mathcal{N}(0, I)$ , ensuring continuity and completeness of the latent space for generative sampling.

[ZH] KL 散度对学习到的潜在分布  $q(z|x)$  进行正则化，使其接近标准高斯先验  $p(z) \sim \mathcal{N}(0, I)$ ，从而确保潜在空间具有连续性和完备性以进行生成采样。

## 第 33 页: GAN 的理想平衡状态 / Ideal Equilibrium of GAN

*In the ideal equilibrium state of GAN training, which statement is the most accurate?*

在 GAN 训练的理想平衡状态下，哪项陈述最准确？

- A. The Discriminator can perfectly distinguish real and fake samples (判别器能完

美区分真实和虚假样本)

- B. The Generator produces samples that are indistinguishable from real data, and the Discriminator outputs approximately 0.5 (生成器生成的样本与真实数据无法区分, 判别器输出约为 0.5)
- C. The Generator no longer needs noise as input (生成器不再需要噪声作为输入)
- D. The Discriminator is removed after training starts (训练开始后判别器被移除)

答案 / Answer: **B**

解析 / Explanation:

[EN] At equilibrium, the generator produces realistic samples that fool the discriminator, and the discriminator can no longer differentiate real from fake, thus outputting 0.5 (random guessing).

[ZH] 在平衡状态下, 生成器能生成欺骗判别器的真实样本, 而判别器无法再区分真实与虚假样本, 因此输出 0.5 (随机猜测)。

## 第 62 页: 多噪声尺度的必要性 / Need for Multiple Noise Levels in Score-Based Models

*In score-based generative models, why do we use multiple noise levels  $\sigma_1 < \sigma_2 < \dots < \sigma_L$  instead of only one fixed noise scale?*

在基于分数的生成模型中, 为什么我们使用多个噪声水平  $\sigma_1 < \sigma_2 < \dots < \sigma_L$  而不是单一的固定噪声尺度?

- A. Because a single noise scale cannot train a neural network (单一噪声尺度无法训练神经网络)
- B. Because larger noise always produces sharper images (较大的噪声总是产生更清晰的图像)
- C. Because the number of noise levels must equal the number of model parameters (噪声水平的数量必须等于模型参数的数量)
- D. Because small noise preserves data details, while large noise covers low-density regions; multiple scales balance both (小噪声保留数据细节, 大噪声覆盖低密度区域; 多尺度平衡两者)



答案 / Answer: **D**

解析 / Explanation:

[EN] Small noise preserves fine-grained details but fails in low-density regions; large noise populates low-density regions but corrupts data. Multiple noise scales enable accurate score estimation across all regions via annealed Langevin dynamics.

[ZH] 小噪声保留细节但在低密度区域失效；大噪声覆盖低密度区域但会破坏数据。多噪声尺度通过退火朗之万动力学在所有区域实现准确的分数的估计。

## 第 77 页: 无分类器引导 / Classifier-Free Guidance

*Which statement about Classifier-Free Guidance is correct?*

关于无分类器引导的陈述哪项是正确的？

- A. It requires an additional classifier to provide gradients (需要额外的分类器提供梯度)
- B. It does not use conditional information at all (完全不使用条件信息)
- C. It randomly drops the condition during training, allowing the same model to learn both conditional and unconditional generation (它在训练期间随机丢弃条件, 使同一模型能够同时学习条件和无条件生成)
- D. It can only be used for unconditional image generation (只能用于无条件图像生成)

答案 / Answer: **C**

解析 / Explanation:

[EN] Classifier-free guidance uses conditioning dropout (e.g., 10-20% of the time, condition  $y$  is removed), enabling a single model to learn both  $p(x|y)$  and  $p(x)$ . Sampling then combines conditional and unconditional scores.

[ZH] 无分类器引导使用条件丢弃（例如，10-20% 的时间条件  $y$  被移除），使单一模型能够同时学习  $p(x|y)$  和  $p(x)$ 。采样时结合条件和无条件分数。

## 第 92 页: 模型与关键思想匹配 / Match Model with Key Idea

Match each generative model with its key idea.

将每个生成模型与其关键思想进行匹配。

| Model                   | Key Idea                                                                    |
|-------------------------|-----------------------------------------------------------------------------|
| AE                      | B. Uses an encoder-decoder structure to minimize reconstruction error       |
| VAE                     | C. Encodes inputs as distributions and regularizes the latent space with KL |
| GAN                     | A. Uses adversarial training to make generated samples fool a discriminator |
| Score / Diffusion Model | D. Learns a score function or denoising process to gradually generate data  |

### 解析 / Explanation:

[EN] AEs minimize reconstruction error; VAEs add KL regularization; GANs use adversarial game; Score/Diffusion models learn gradients of log-density or denoising process.

[ZH] 自编码器最小化重建误差；变分自编码器添加 KL 正则化；生成对抗网络使用对抗博弈；分数/扩散模型学习对数密度的梯度或去噪过程。

## Week13-Vision-Language Models

### 第?? 页: 对比学习的核心思想 (补充)

What is the core learning objective of CLIP during pre-training?

CLIP 在预训练期间的核心学习目标是什么？

答案 / Answer: **Contrastive learning: maximize cosine similarity for matched image-text pairs while minimizing it for unmatched pairs.** / **对比学习：最大化匹配的图像-文本对的余弦相似度，同时最小化不匹配对的相似度。**

### 解析 / Explanation:

[EN] CLIP uses a contrastive loss (InfoNCE) trained on 400 million image-text pairs to align image and text embeddings in a shared space.

[ZH] CLIP 使用对比损失 (InfoNCE) 在 4 亿图像-文本对上训练，以在共享空间中对齐图像和文本嵌入。

### 第?? 页: 零样本推理的实现 (补充)

How does CLIP perform zero-shot image classification?

CLIP 如何执行零样本图像分类？

**答案 / Answer:** Reformats class names into sentences (e.g., "a photo of a dog"), computes text embeddings as classification weights, and compares with image embeddings via cosine similarity. / 将类别名称重写为句子（如 "a photo of a dog"），计算文本嵌入作为分类权重，通过余弦相似度与图像嵌入进行比较。

**解析 / Explanation:**

[EN] This avoids task-specific training and enables generalization to unseen classes by leveraging the alignment learned during pre-training.

[ZH] 这避免了特定任务的训练，并利用预训练期间学习到的对齐能力实现对未见类别的泛化。

## 第?? 页: CoOp vs CoCoOp (补充)

*What is the main limitation of CoOp that CoCoOp addresses?*

CoCoOp 解决了 CoOp 的什么主要局限性？

**答案 / Answer:** CoOp learns a static context that does not generalize well to unseen classes; CoCoOp generates input-conditional prompts for better generalization. / CoOp 学习静态上下文，难以泛化到未见类别；CoCoOp 生成输入条件化的提示以获得更好的泛化能力。

**解析 / Explanation:**

[EN] CoCoOp uses a lightweight network to condition the prompt on each image instance, making it dynamic and more adaptable.

[ZH] CoCoOp 使用轻量级网络使提示条件化于每个图像实例，使其具有动态性和更强的适应性。

## 第?? 页: TaskRes 的核心思想 (补充)

*What is the key idea of Task Residual Tuning (TaskRes)?*

任务残差调优的核心思想是什么？

**答案 / Answer:** Explicitly decouples prior knowledge from pre-trained VLMs and new knowledge for the target task by operating directly on the text-based

classifier. / 通过在基于文本的分类器上直接操作，显式解耦预训练 VLM 的先验知识和目标任务的新知识。

解析 / Explanation:

[EN] TaskRes balances old knowledge inheritance and task-specific exploration, avoiding the pitfalls of prompt tuning (loss of prior) and adapter tuning (limited flexibility).

[ZH] TaskRes 平衡了旧知识继承和任务特定探索，避免了提示调优（先验丢失）和适配器调优（灵活性有限）的缺陷。

## 第?? 页: TCM 在文本检测中的应用 (补充)

*How does TCM (Turning CLIP Model) adapt CLIP for scene text detection?*

TCM 如何使 CLIP 适配场景文本检测任务?

答案 / Answer: **Designs cross-modal interaction via visual prompt learning to recover locality features, using cross-attention to model interactions between image and text embeddings, and outputs a binary text segmentation map.** / 通过视觉提示学习设计跨模态交互以恢复局部特征，使用交叉注意力建模图像和文本嵌入之间的交互，并输出二值文本分割图。

解析 / Explanation:

[EN] TCM leverages CLIP's pre-trained knowledge without additional pre-training, making it effective for text detection with limited labeled data.

[ZH] TCM 利用 CLIP 的预训练知识，无需额外的预训练过程，使其在标注数据有限的情况下对文本检测有效。

## 第?? 页: 提升 VLM 计数能力的方法 (补充)

*What approach does "Teaching CLIP to Count" use to improve counting ability?*

“Teaching CLIP to Count” 使用什么方法来提升计数能力?

答案 / Answer: **Creates a specialized counting dataset with images and captions describing object counts, then fine-tunes CLIP with an additional count-based contrastive loss  $L = L_{CLIP} + \lambda L_{count}$ .** / 创建专门的计数数据集，包含描述物体数量的图像和标题，然后使用额外的基于计数的对比损失  $L = L_{CLIP} + \lambda L_{count}$  对 CLIP 进行微调。

### 解析 / Explanation:

[EN] This addresses CLIP's weakness in compositional understanding (e.g., counting) by augmenting the training data and loss function.

[ZH] 这通过增强训练数据和损失函数来解决 CLIP 在组合性理解(如计数)方面的弱点。

## 第?? 页: 描述性分类的核心公式 (补充)

*What is the scoring function used in descriptive classification?*

描述性分类中使用什么评分函数?

**答案 / Answer:**  $s(c, x) = \frac{1}{|D(c)|} \sum_{d \in D(c)} \phi(d, x)$ , where  $D(c)$  are descriptors for category  $c$  and  $\phi(d, x)$  is the log probability that descriptor  $d$  pertains to image  $x$ . /  $s(c, x) = \frac{1}{|D(c)|} \sum_{d \in D(c)} \phi(d, x)$ , 其中  $D(c)$  是类别  $c$  的描述符集合,  $\phi(d, x)$  是描述符  $d$  与图像  $x$  相关的对数概率。

### 解析 / Explanation:

[EN] This aggregates evidence from multiple visual descriptors (e.g., "stripes", "claws" for tiger) rather than relying solely on category names, improving interpretability and robustness.

[ZH] 这聚合了来自多个视觉描述符的证据(例如,老虎的“条纹”、“爪子”),而不是仅依赖类别名称,从而提高了可解释性和鲁棒性。